

Logic Families

Two basic techniques are used for the manufacture of ICs, Bipolar and metal-oxide-semiconductor (MOS) technologies. Bipolar technology is preferred for SSI and MSI, because it is faster.

MOS technology is preferred for LSI because of increased density of MOSFET in the same chip area.

- (i) RTL (Resistor-transistor logic)
- (ii) DTL (Diode-transistor logic)
- (iii) TTL (T^2L) (Transistor-transistor logic)
- (iv) I²L (Integrated-injected logic)
- (v) ECL (Emitter coupled logic)
- (vi) MOS (Metal-oxide semiconductor)
- (vii) CMOS (Complementary metal-oxide-semiconductor)

Sometimes speed may be the main requirement whereas another time minimum power dissipation might be main consideration. The different characteristics of the logic family are as under:-

(i) Propagation time delay:- This is the average transition delay time for the signal to propagate from input to output when the signal changes its value. This determines how fast is the no. of gates that can be driven by it.

(ii) Fan-in and Fan-out:- The maximum no. of inputs which can be applied to a logic gate is known as Fan-in. Fan-out of a logic gate is the no. of gates that can be driven by it.

(iii) Noise Margin:- The maximum value of noise signal that a system can reject with performance unaffected.

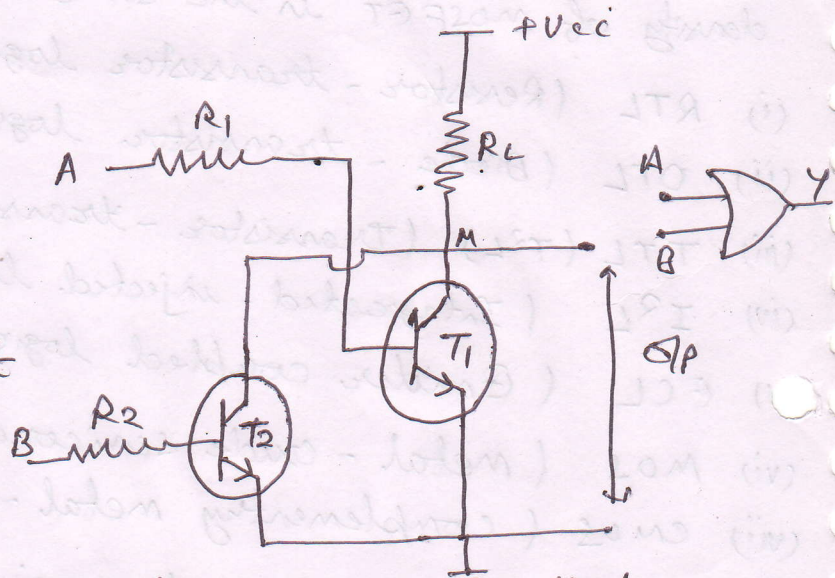
(iv) Noise immunity:- It is the voltage which when applied as the input of a logic gate will cause the o/p to change the o/p state for a given supply voltage.

(v) Power dissipation:- The power consumed by the gate, which must be available from the power supply.

Register-Transistor Logic (RTL)

It considers only, transistor and resistor for logic. The transistor can be connected in series or parallel to perform logic operation.

When input A to transistor T_1 is high (at $+V_{cc}$), it will conduct thereby reducing the voltage of point M to almost 0V. i.e. o/p will be at logic 0. Similarly for B to transistor T_2 is high.



If either or both input are high, the o/p will be 0. When both input are low, the T_1 and T_2 will be cut-off (OFF), hence point M goes to $+V_{cc}$ so o/p will be high (logic 1).

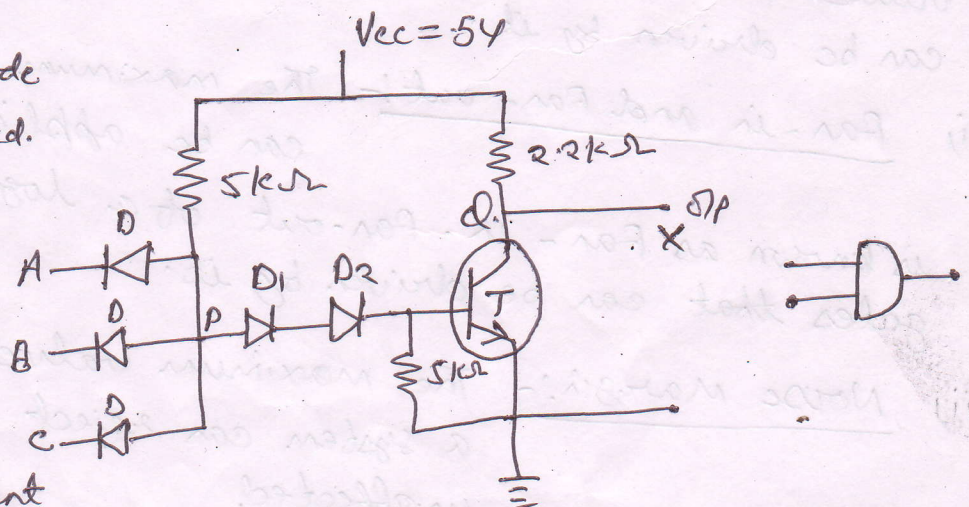
Diode-Transistor Logic (DTL)

It consists of diodes, resistors and transistor.

If any i/p is low diode will be forward biased. i.e. diode conduct. and voltage at point P becomes low.

i.e. diode D_1 and D_2 do not conduct. and the base current of transistor T becomes zero.

The transistor is OFF. This makes the o/p of transistor T high i.e. the o/p X is high (logic 1).



When all i/p are high (logic 1), the diodes D are cut off. The voltage at P. rises toward V_{cc} which results a base current to transistor T. When base current is large, the transistor is driven to saturation. Now the Q becomes low i.e. o/p x falls to low (logic 0).

Transistor-Transistor Logic

It uses multiple emitter transistor because to replace the input diodes of DTL.

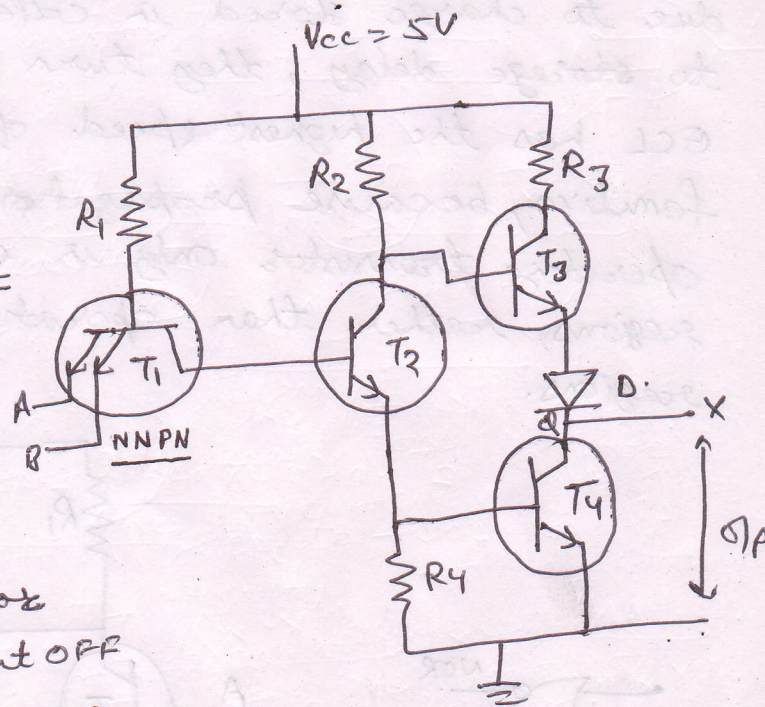
The input A and B are high.

The transistor T_1 has no emitter current because emitter base junction is reversed bias. Hence it is OFF. The base collector junction of T_1 is forward biased and supply base

current to T_2 . Hence T_2 is turned ON. Now its collector potential goes low which cuts off the transistor T_3 . At the same

time, T_2 feeds base current to T_4 . i.e. T_4 is turned ON. i.e. o/p will be low (logic 0). Hence, when the inputs are high (logic 1) the output is low (logic 0).

Similarly, with any input low (logic 0), the output is high (logic 1).



(NAND Gate)

Emitter-Coupled Logic (ECL)

Current Mode Logic (CML)

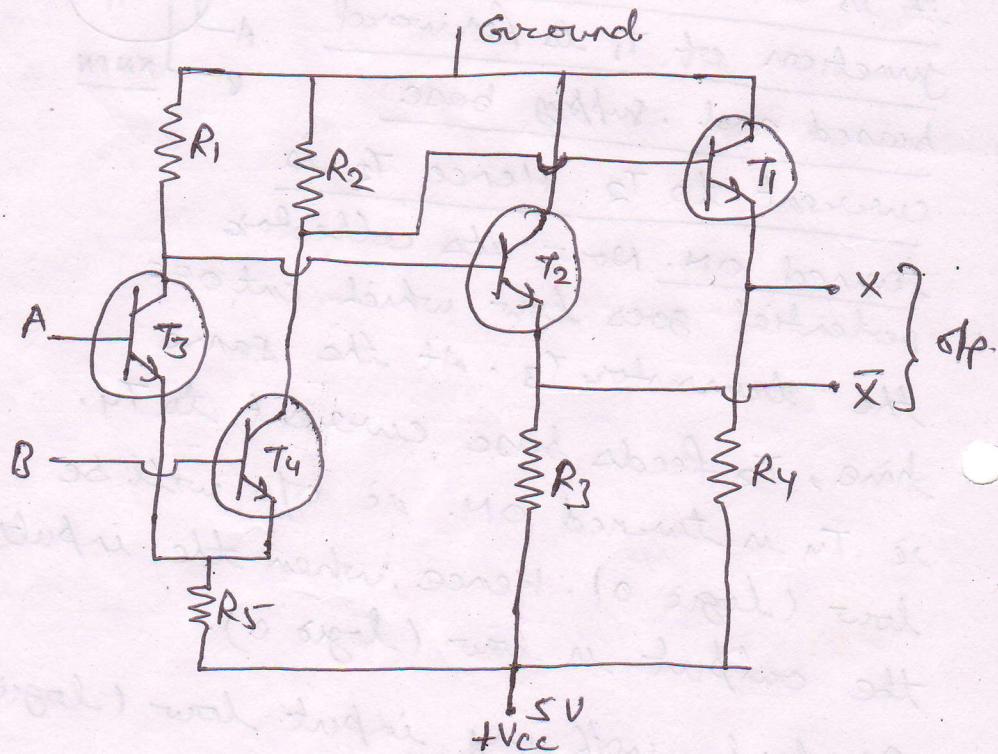
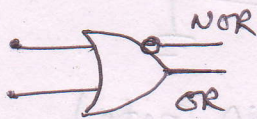
Current Steering Logic (CSL)

Non-saturated logic (NSL)

(i.e. Because transistors never operate fully saturated or cut-off)

When transistors are operated in saturation condition, then due to charge stored in collector and base region, i.e. due to storage delay, they turn ON or OFF slowly.

ECL has the highest speed of any of the available logic families; because propagation delay is eliminated the operating transistor only is either active or the cut-off regions rather than operating in saturation or cut-off regions.



Integrated-Injection Logic (IIL or I^2L)

I^2L do not contain resistors. Hence they are smaller in size. I^2L circuits have less power dissipation and hence they are used in large computers.

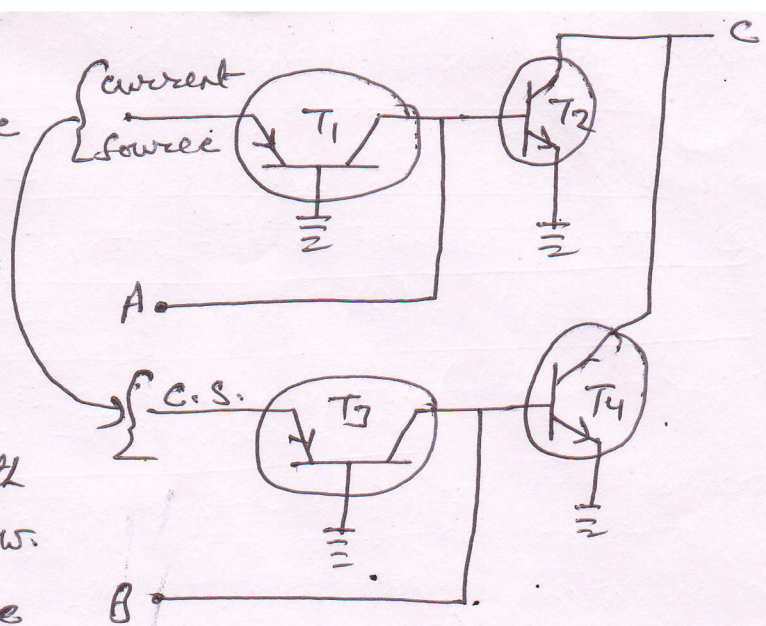
When A is low, then the current to the base of T_2 will be shorted to ground i.e. $T_2 = \text{OFF}$.

The i/p B controls T_4 in similar way. So the o/p = high only when both input A and B are low.

When A is high the base current flowing to T_2 will turn it ON. i.e. output

C low. Similarly, the o/p is low when B is high.

thus o/p would be low when either A or B or both are high.



NOR gate